WIKIPEDIA
The Free Encyclopedia

Potato

The **potato** (/pəˈteɪtoʊ/) is a starchy root vegetable native to the Americas that is consumed as a staple food in many parts of the world. Potatoes are tubers of the plant ***Solanum tuberosum***, a perennial in the nightshade family Solanaceae.

Wild potato species can be found from the southern United States to southern Chile. Genetic studies show that the cultivated potato has a single origin, in the area of present-day southern Peru and extreme northwestern Bolivia. Potatoes were domesticated there about 7,000–10,000 years ago from a species in the *S. brevicaule* complex. Many varieties of the potato are cultivated in the Andes region of South America, where the species is indigenous.

The Spanish introduced potatoes to Europe in the second half of the 16th century from the Americas. They are a staple food in many parts of the world and an integral part of much of the world's food supply. Following millennia of selective breeding, there are now over 5,000 different varieties of potatoes. The potato remains an essential crop in Europe, especially Northern and Eastern Europe, where per capita production is still the highest in the world, while the most rapid expansion in production during the 21st century was in southern and eastern Asia, with China and India leading the world production as of 2021.

Like the tomato and the nightshades, the potato is in the genus *Solanum*; the aerial parts of the potato contain the toxin solanine. Normal potato tubers that have been grown and stored properly produce glycoalkaloids in negligible amounts, but, if sprouts and potato skins are exposed to light, tubers can become toxic.

Potato	
Potato cultivars appear in a variety of colors, shapes, and sizes.	
Scientific classification	
Kingdom:	Plantae
Clade:	Tracheophytes
Clade:	Angiosperms
Clade:	Eudicots
Clade:	Asterids
Order:	Solanales
Family:	Solanaceae
Genus:	*Solanum*
Species:	***S. tuberosum***
Binomial name	
Solanum tuberosum L.	
Synonyms	
see list	

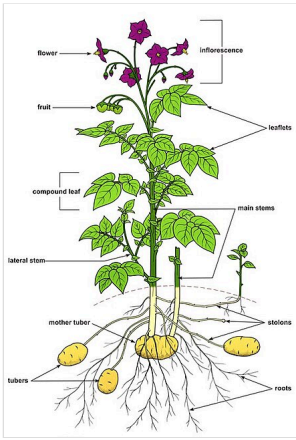
Etymology

The English word "potato" comes from Spanish *patata*, in turn from Taíno *batata*, which means "sweet potato", not the plant now known as simply "potato".[1]

The name "spud" for a potato is from the 15th century *spudde*, a short knife or dagger, probably related to Danish *spyd*, "spear". From around 1840, the name transferred to the tuber itself.[2]

At least six languages—Afrikaans, Dutch, French, (West) Frisian, Hebrew, Persian[3] and some variants of German—use a term for "potato" that means "earth apple" or "ground apple".[4][5]

Description



Morphology of the potato plant; tubers are forming from stolons.

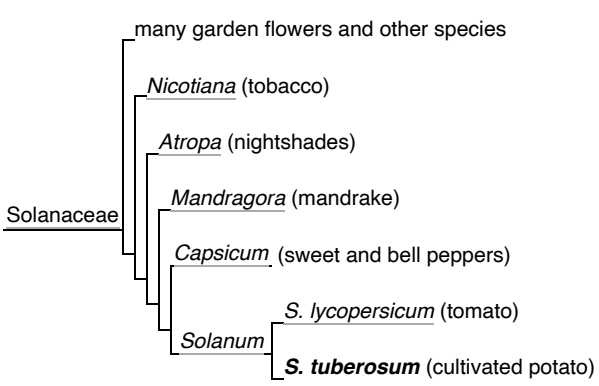
Potato plants are herbaceous perennials that grow up to 1 metre (3.3 ft) high. The stems are hairy. The leaves have roughly four pairs of leaflets. The flowers range from white or pink to blue or purple; they are yellow at the centre, and are insect-pollinated.[6]

The plant develops tubers to store nutrients. These are not roots but stems that form from thickened rhizomes at the tips of long thin stolons. On the surface of the tubers there are "eyes," which act as sinks to protect the vegetative buds from which the stems originate. The "eyes" are arranged in helical form. In addition, the tubers have small holes that allow breathing, called lenticels. The lenticels are circular and their number varies depending on the size of the tuber and environmental conditions.[7] Tubers form in response to decreasing day length, although this tendency has been minimized in commercial varieties.[8]

After flowering, potato plants produce small green fruits that resemble green cherry tomatoes, each containing about 300 very small seeds.[9]

Phylogeny

Wild and cultivated potatoes, like the tomato, belong to the genus *Solanum*, which is a member of the nightshade family, the Solanaceae. That is a diverse family of flowering plants, often poisonous, that includes the mandrake (*Mandragora*), deadly nightshade (*Atropa*), and tobacco (*Nicotiana*), as shown in the outline phylogenetic tree (many branches omitted). The most commonly cultivated potato is *S. tuberosum*; there are several other species.[10]



A variety of *S. tuberosum tuberosum*, the Chilean potato

The major species grown worldwide is *S. tuberosum* (a tetraploid with 48 chromosomes), and modern varieties of this species are the most widely cultivated. There are also four diploid species (with 24 chromosomes): *S. stenotomum*, *S. phureja*, *S. goniocalyx*, and *S. ajanhuiri*. There are two triploid species (with 36 chromosomes): *S. chaucha* and *S. juzepczukii*. There is one pentaploid cultivated species (with 60 chromosomes): *S. curtilobum*.[11]

There are two major subspecies of *S. tuberosum*.[11] The Andean potato, *S. tuberosum andigena*, is adapted to the short-day conditions prevalent in the mountainous equatorial and tropical regions where it originated. The Chilean potato *S. tuberosum tuberosum*, native to the Chiloé Archipelago, is in contrast adapted to the long-day conditions prevalent in the higher latitude region of southern Chile.[12]

History

Domestication

Wild potato species occur from the southern United States to southern Chile.[13] The potato was first domesticated in southern Peru and northwestern Bolivia[14] by pre-Columbian farmers, around Lake Titicaca.[15] Potatoes were domesticated there about 7,000–10,000 years ago from a species in the *S. brevicaule* complex.[14][15][16]

The earliest archaeologically verified potato tuber remains have been found at the coastal site of Ancon (central Peru), dating to 2500 BC.[17][18] The most widely cultivated variety, *Solanum tuberosum tuberosum*, is indigenous to the Chiloé Archipelago, and has been cultivated by the local indigenous people since before the Spanish conquest.[12][19]

Spread

Following the Spanish conquest of the Inca Empire, the Spanish introduced the potato to Europe in the second half of the 16th century as part of the Columbian exchange. The staple was subsequently conveyed by European mariners (possibly including the Russian-American Company) to territories and ports throughout the world, especially their colonies.[20] European and colonial farmers were slow to adopt farming potatoes, but after 1750 they became an important food staple and field crop[20] and played a major role in the European 19th century population boom.[16] According to conservative estimates, the introduction of the potato was responsible for a quarter of the growth in Old World population and urbanization between 1700 and 1900.[21] However, lack of genetic diversity, due to the very limited number of varieties initially introduced, left the crop vulnerable to disease. In 1845, a plant disease known as late blight, caused by the fungus-like oomycete *Phytophthora infestans*, spread rapidly through the poorer communities of western Ireland as well as parts of the Scottish Highlands, resulting in the crop failures that led to the Great Irish Famine.[22][20]

The International Potato Center, based in Lima, Peru, holds 4,870 types of potato germplasm, most of which are traditional landrace cultivars.[23] In 2009 a draft sequence of the potato genome was made, containing 12 chromosomes and 860 million base pairs, making it a medium-sized plant genome.[24]

It had been thought that most potato cultivars derived from a single origin in southern Peru and extreme Northwestern Bolivia, from a species in the *S. brevicaule* complex.[14][15][16] DNA analysis however shows that more than 99% of all current varieties of potatoes are direct descendants of a subspecies that once grew in the lowlands of south-central Chile.[25]

Most modern potatoes grown in North America arrived through European settlement and not independently from the South American sources. At least one wild potato species, *S. fendleri*, occurs in North America; it is used in breeding for resistance to a nematode species that attacks cultivated potatoes. A secondary center of genetic variability of the potato is Mexico, where important wild species that have been used extensively in modern breeding are found, such as the hexaploid *S. demissum*, used as a source of resistance to the devastating late blight disease (*Phytophthora infestans*).[22] Another relative native to this region, *Solanum bulbocastanum*, has been used to genetically engineer the potato to resist potato blight.[26] Many such wild relatives are useful for breeding resistance to *P. infestans*.[27]

Little of the diversity found in *Solanum* ancestral and wild relatives is found outside the original South American range.[28] This makes these South American species highly valuable in breeding.[28] The importance of the potato to humanity is recognised in the United Nations International Day of Potato, to be celebrated on 30 May each year, starting in 2024.[29]

Breeding

Potatoes, both *S. tuberosum* and most of its wild relatives, are self-incompatible: they bear no useful fruit when self-pollinated. This trait is problematic for crop breeding, as all sexually-produced plants must be hybrids. The gene responsible for self-incompatibility, as well as mutations to disable it, are now known. Self-compatibility has successfully been introduced both to diploid potatoes (including a special line of *S. tuberosum*) by CRISPR-Cas9.[30] Plants having a 'Sli' gene produce pollen which is compatible to its own parent and plants with similar S genes.[31] This gene was cloned by Wageningen University and Solynta in 2021, which would allow for faster and more focused breeding.[30][32]

Diploid hybrid potato breeding is a recent area of potato genetics supported by the finding that simultaneous homozygosity and fixation of donor alleles is possible.[33] Wild potato species useful for breeding blight resistance include *Solanum desmissum* and *S. stoloniferum*, among others.[34]

Varieties

Potatoes of different colors

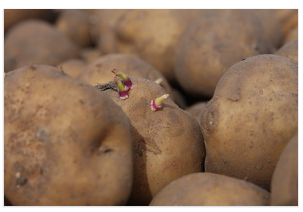
There are some 5,000 potato varieties worldwide, 3,000 of them in the Andes alone — mainly in Peru, Bolivia, Ecuador, Chile, and Colombia. Over 100 cultivars might be found in a single valley, and a dozen or more might be maintained by a single agricultural household.[35][36] The European Cultivated Potato Database is an online collaborative database of potato variety descriptions updated and maintained by the Scottish Agricultural Science Agency within the framework of the European Cooperative Programme for Crop Genetic Resources Networks—which is run by the International Plant Genetic Resources Institute.[37] Around 80 varieties are commercially available in the UK.[38]

For culinary purposes, varieties are often differentiated by their waxiness: floury or mealy *baking* potatoes have more starch (20–22%) than waxy *boiling* potatoes (16–18%). The distinction may also arise from variation in the comparative ratio of two different potato starch compounds: amylose and amylopectin. Amylose, a long-chain molecule, diffuses from the starch granule when cooked in water, and lends itself to dishes where the potato is mashed. Varieties that contain a slightly higher amylopectin content, which is a highly branched molecule, help the potato retain its shape after being boiled in water.[39] Potatoes that are good for making potato chips or potato crisps are sometimes called "chipping potatoes", which means they meet the basic requirements of similar varietal characteristics, being firm, fairly clean, and fairly well-shaped.[40]

Immature potatoes may be sold fresh from the field as "creamer" or "new" potatoes and are particularly valued for their taste. They are typically small in size and tender, with a loose skin, and flesh containing a lower level of starch than other potatoes. In the United States they are generally either a Yukon Gold potato or a red potato, called gold creamers or red creamers respectively.[41][42] In the UK, the Jersey Royal is a famous type of new potato.[43]

Dozens of potato cultivars have been selectively bred specifically for their skin or flesh color, including gold, red, and blue varieties.[44] These contain varying amounts of phytochemicals, including carotenoids for gold/yellow or polyphenols for red or blue cultivars.[45] Carotenoid compounds include provitamin A alpha-carotene and beta-carotene, which are converted to the essential nutrient, vitamin A, during digestion. Anthocyanins mainly responsible for red or blue pigmentation in potato cultivars do not have nutritional significance, but are used for visual variety and consumer appeal.[46] In 2010, potatoes were bioengineered specifically for these pigmentation traits.[47]

Genetic engineering



Amflora potatoes, modified to produce pure amylopectin starch

Genetic research has produced several genetically modified varieties. 'New Leaf', owned by Monsanto Company, incorporates genes from *Bacillus thuringiensis* (source of most *Bt* toxins in transcrop use), which confers resistance to the Colorado potato beetle; 'New Leaf Plus' and 'New Leaf Y', approved by US regulatory agencies during the 1990s, also include resistance to viruses. McDonald's, Burger King, Frito-Lay, and Procter & Gamble announced they would not use genetically modified potatoes, and Monsanto published its intent to discontinue the line in March 2001.[48]

Potato starch contains two types of glucan, amylose and amylopectin, the latter of which is most industrially useful. Waxy potato varieties produce waxy potato starch, which is almost entirely amylopectin, with little or no amylose. BASF developed the 'Amflora' potato, which was modified to express antisense RNA to inactivate the gene for granule bound starch synthase, an enzyme which catalyzes the formation of amylose.[49] 'Amflora' potatoes therefore produce starch consisting almost entirely of amylopectin, and are thus more useful for the starch industry. In 2010, the European Commission cleared the way for 'Amflora' to be grown in the European Union for industrial purposes only—not for food. Nevertheless, under EU rules, individual countries have the right to decide whether they will allow this potato to be grown on their territory. Commercial planting of 'Amflora' was expected in the Czech Republic and Germany in the spring of 2010, and Sweden and the Netherlands in subsequent years.[50]

The 'Fortuna' GM potato variety developed by BASF was made resistant to late blight by introgressing two resistance genes, blb1 and blb2, from *S. bulbocastanum*, a wild potato native to Mexico.[51][52][53] Rpi-blb1 is a nucleotide-binding leucine-rich repeat (NB-LRR/NLR), an R-gene-produced immunoreceptor.[51]

In October 2011 BASF requested cultivation and marketing approval as a feed and food from the EFSA. In 2012, GMO development in Europe was stopped by BASF.[54][55] In November 2014, the United States Department of Agriculture (USDA) approved a genetically modified potato developed by Simplot, which contains genetic modifications that prevent bruising and produce less acrylamide when fried than conventional potatoes; the modifications do not cause new proteins to be made, but rather prevent proteins from being made via RNA interference.[56]

Genetically modified varieties have met public resistance in the U.S. and in the European Union.[57][58]

Cultivation

Seed potatoes

Potatoes are generally grown from "seed potatoes", tubers specifically grown to be free from disease and to provide consistent and healthy plants. To be disease free, the areas where seed potatoes are grown are selected with care. In the US, this restricts production of seed potatoes to only 15 states out of all 50 states where potatoes are grown. These locations are selected for their cold, hard winters that kill pests and summers with long sunshine hours for optimum growth.[59] In the UK, most seed potatoes originate in Scotland, in areas where westerly winds reduce aphid attacks and the spread of potato virus pathogens.[60]

Phases of growth

Potato growth can be divided into five phases. During the first phase, sprouts emerge from the seed potatoes and root growth begins. During the second, photosynthesis begins as the plant develops leaves and branches above-ground and stolons develop from lower leaf axils on the below-ground stem. In the third phase the tips of the stolons swell, forming new tubers, and the shoots continue to grow, with flowers typically developing soon after. Tuber bulking occurs during the fourth phase, when the plant begins investing the majority of its resources in its newly formed tubers. At this phase, several factors are critical to a good yield: optimal soil moisture and temperature, soil nutrient availability and balance, and resistance to pest attacks. The fifth phase is the maturation of the tubers: the leaves and stems senesce and the tuber skins harden.[61][62]

New tubers may start growing at the surface of the soil. Since exposure to light leads to an undesirable greening of the skins and the development of solanine as a protection from the sun's rays, growers cover surface tubers. Commercial growers cover them by piling additional soil around the base of the plant as it grows (called "hilling" up, or in British English "earthing up"). An alternative method, used by home gardeners and smaller-scale growers, involves covering the growing area with mulches such as straw or plastic sheets.[63]

At farm scale, potatoes require a well-drained neutral or mildly acidic soil (pH 6 or 7) such as a sandy loam. The soil is prepared using deep tillage, for example with a chisel plow or ripper. In areas where irrigation is needed, the field is leveled using a landplane so that water can be supplied evenly. Manure can be added after initial irrigation; the soil is then broken up with a disc harrow. The potatoes are planted using a potato planter machine in rows 80 centimetres (31 in) apart.[64] At garden scale, potatoes are planted in trenches or individual holes some 15 centimetres (5.9 in) deep in soil, preferably with additional organic matter such as garden compost or manure. Alternatively, they can be planted in containers or bags filled with a free-draining compost.[65] Potatoes are sensitive to heavy frosts, which damage them in the ground or when stored.[66]

Planting

Field in Fort Fairfield, Maine

Immature potato plants

Potatoes grown in a tall bag are common in gardens as they minimize digging.

Pests and diseases

Late blight

The historically significant *Phytophthora infestans*, the cause of late blight, remains an ongoing problem in Europe[22] and the United States.[67] Other potato diseases include *Rhizoctonia*, *Sclerotinia*, *Pectobacterium carotovorum* (black leg), powdery mildew, powdery scab and leafroll virus.[68][69]

Insects that commonly transmit potato diseases or damage the plants include the Colorado potato beetle, the potato tuber moth, the green peach aphid (*Myzus persicae*), the potato aphid, *Tuta absoluta*, beet leafhoppers, thrips, and mites. The Colorado potato beetle is considered the most important insect defoliator of potatoes, devastating entire crops.[70] The potato cyst nematode is a microscopic worm that feeds on the roots, thus causing the potato plants to wilt. Since its eggs can survive in the soil for several years, crop rotation is recommended.[71]

Harvest

A modern potato harvester

On a small scale, potatoes can be harvested using a hoe or spade, or simply by hand. Commercial harvesting is done with large potato harvesters, which scoop up the plant and surrounding earth. This is transported up an apron chain consisting of steel links several feet wide, which separates some of the earth. The chain deposits into an area where further separation occurs. The most complex designs use vine choppers and shakers, along with a blower system to separate the potatoes from the plant. The result is then usually run past workers who continue to sort out plant material, stones, and rotten potatoes before the potatoes are continuously delivered to a wagon or truck. Further inspection and separation occurs when the potatoes are unloaded from the field vehicles and put into storage.[72]

Potatoes are usually cured after harvest to improve skin-set. Skin-set is the process by which the skin of the potato becomes resistant to skinning damage. Potato tubers may be susceptible to skinning at harvest and suffer skinning damage during harvest and handling operations. Curing allows the skin to fully set and any wounds to heal. Wound-healing prevents infection and water-loss from the tubers during storage. Curing is normally done at relatively warm temperatures (10 to 16 °C or 50 to 60 °F) with high humidity and good gas-exchange if at all possible.[73]

Storage

Transporting to cold storage in India

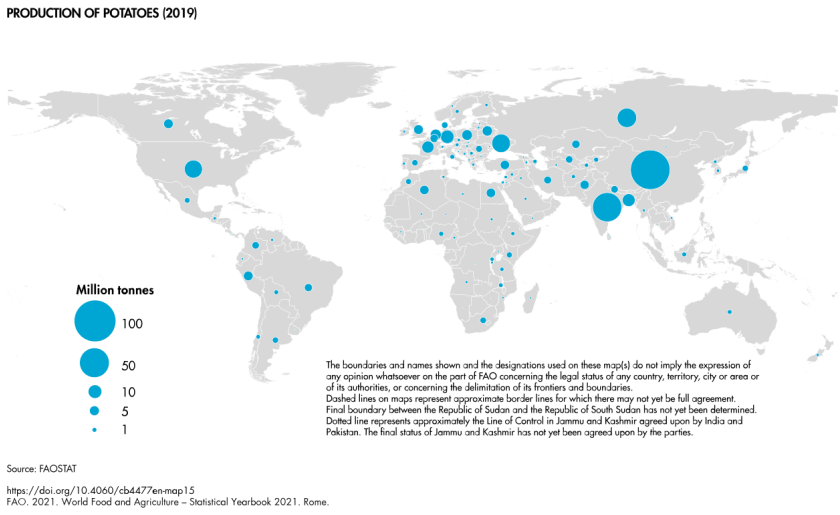
Storage facilities need to be carefully designed to keep the potatoes alive and slow the natural process of sprouting which involves the breakdown of starch. It is crucial that the storage area be dark, ventilated well, and, for long-term storage, maintained at temperatures near 4 °C (39 °F). For short-term storage, temperatures of about 7 to 10 °C (45 to 50 °F) are preferred.[74]

Temperatures below 4 °C (39 °F) convert the starch in potatoes into sugar, which alters their taste and cooking qualities and leads to higher acrylamide levels in the cooked product, especially in deep-fried dishes. The discovery of acrylamides in starchy foods in 2002 has caused concern, but it is not likely that the acrylamides in food, even if it is somewhat burnt, causes cancer in humans.[75]

Chemicals are used to suppress sprouting of tubers during storage. Chlorpropham is the main chemical used, but it has been banned in the EU over toxicity concerns.[76] Alternatives include ethylene, spearmint and orange oils, and 1,4-dimethylnaphthalene.[76]

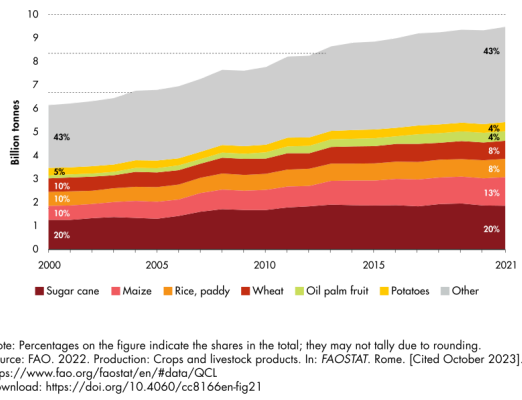
Under optimum conditions in commercial warehouses, potatoes can be stored for up to 10–12 months.[74] The commercial storage and retrieval of potatoes involves several phases: first *drying* surface moisture; *wound healing* at 85% to 95% relative humidity and temperatures below 25 °C (77 °F); a staged *cooling phase*; a *holding* phase; and a *reconditioning* phase, during which the tubers are slowly warmed. Mechanical ventilation is used at various points during the process to prevent condensation and the accumulation of carbon dioxide.[74]

Production



Potato production – 2021	
Country	**Production (millions of tonnes)**
China	94.3
India	54.2
Ukraine	21.4
United States	18.6
Russia	18.3
World	**376**
Source: FAOSTAT of the United Nations[77]	

Production of potatoes (2019)[78]



Potatoes are one of the most widely produced primary crops in the world.

In 2021, world production of potatoes was 376 million tonnes (370,000,000 long tons; 414,000,000 short tons), led by China with 25% of the total. Other major producers were India and Ukraine (table).

The world dedicated 18.6 million hectares (46 million acres) to potato cultivation in 2010; the world average yield was 17.4 tonnes per hectare (7.8 short tons per acre). The United States was the most productive country, with a nationwide average yield of 44.3 tonnes per hectare (19.8 short tons per acre).[79]

New Zealand farmers have demonstrated some of the best commercial yields in the world, ranging between 60 and 80 tonnes per hectare, some reporting yields of 88 tonnes of potatoes per hectare.[80][81][82]

There is a big gap among various countries between high and low yields, even with the same variety of potato. Average potato yields in developed economies ranges between 38 and 44 metric tons per hectare (15 and 18 long ton/acre; 17 and 20 short ton/acre). China and India accounted for over a third of world's production in 2010, and had yields of 14.7 and 19.9 metric tons per hectare (5.9 and 7.9 long ton/acre; 6.6 and 8.9 short ton/acre) respectively.[79] The yield gap between farms in developing economies and developed economies represents an opportunity loss of over 400 million metric tons (440 million short tons; 390 million long tons) of potato, or an amount greater than 2010 world potato production. Potato crop yields are determined by factors such as the crop breed, seed age and quality, crop management practices and the plant environment. Improvements in one or more of these yield determinants, and a closure of the yield gap, could be a major boost to food supply and farmer incomes in the developing world.[83][84] The food energy yield of potatoes—about 95 gigajoules per hectare (9.2 million kilocalories per acre)—is higher than that of maize (78 GJ/ha or 7.5 million kcal/acre), rice (77 GJ/ha or 7.4 million kcal/acre), wheat (31 GJ/ha or 3 million kcal/acre), or soybeans (29 GJ/ha or 2.8 million kcal/acre).[85]

Impact of climate change on production

Climate change is predicted to have significant effects on global potato production.[86] Like many crops, potatoes are likely to be affected by changes in atmospheric carbon dioxide, temperature and precipitation, as well as interactions between these factors.[86] As well as affecting potatoes directly, climate change will also affect the distributions and populations of many potato diseases and pests. While the potato is less important than maize, rice, wheat and soybeans, which are collectively responsible for around two-thirds of all calories consumed by humans (both directly and indirectly as animal feed),[87] it still is one of the world's most important food crops.[88] Altogether, one 2003 estimate suggests that future (2040–2069) worldwide potato yield would be 18-32% lower than it was at the time, driven by declines in hotter areas like Sub-Saharan Africa,[86] unless farmers and potato cultivars can adapt to the new environment.[89]

Potato plants and crop yields are predicted to benefit from the CO2 fertilization effect,[90] which would increase photosynthetic rates and therefore growth, reduce water consumption through lower transpiration from stomata and increase starch content in the edible tubers.[86] However, potatoes are more sensitive to soil water deficits than some other staple crops like wheat.[91] In the UK, the amount of arable land suitable for rainfed potato production is predicted to decrease by at least 75%.[92] These changes are likely to lead to increased demand for irrigation water, particularly during the potato growing season.[86]

Potatoes grow best under temperate conditions.[93] Temperatures above 30 °C (86 °F) have negative effects on potato crops, from physiological damage such as brown spots on tubers, to slower growth, premature sprouting, and lower starch content.[94] These effects reduce crop yield, affecting both the number and the weight of tubers. As a result, areas where

current temperatures are near the limits of potatoes' temperature range (e.g. much of sub-Saharan Africa)[86] will likely suffer large reductions in potato crop yields in the future.[93] On the other hand, low temperatures reduce potato growth and present risk of frost damage.[86]

Changes in pests and diseases

Plant destroyed by Colorado potato beetle (*Leptinotarsa decemlineata*) larvae

Climate change is predicted to affect many potato pests and diseases. These include:

- Insect pests such as the potato tuber moth and Colorado potato beetle, which are predicted to spread into areas currently too cold for them.[86]
- Aphids which act as vectors for many potato viruses and will spread under increased temperatures.[95]
- Pathogens causing potato blackleg disease (e.g. Dickeya) grow and reproduce faster at higher temperatures.[96]
- Bacterial infections such as Ralstonia solanacearum will benefit from higher temperatures and spread more easily through flash flooding.[86]
- Late blight benefits from higher temperatures and wetter conditions.[97] Late blight is predicted to become a greater threat in some areas (e.g. in Finland)[86] and become a lesser threat in others (e.g. in the United Kingdom).[90]

Adaptation strategies

Potato production is expected to decline in many areas due to hotter temperatures and decreased water availability. Conversely, production is predicted to become possible in high altitude and latitude areas where it has been limited by frost damage, such as in Canada and Russia.[93] This will shift potato production to cooler areas, mitigating much of the projected decline in yield. However, this may trigger competition for land between potato crops and other land uses, mostly due to changes in water and temperature regimes.[93]

The other approach is through the development of varieties or cultivars which would be more adapted to altered conditions. This can be done through 'traditional' plant breeding techniques and genetic modification. These techniques allow for the selection of specific traits as a new cultivar is developed. Certain traits, such as heat stress tolerance, drought tolerance, fast growth/early maturation and disease resistance, may play an important role in creating new cultivars able to maintain yields under stressors induced by climate change.[94]

For instance, developing cultivars with greater heat stress tolerance would be critical for maintaining yields in countries with potato production areas near current cultivars' maximum temperature limits (e.g. Sub-Saharan Africa, India).[98] Superior drought resistance can be achieved through improved water use efficiency (amount of food produced per amount of water used) or the ability to recover from short drought periods and still produce acceptable yields. Further, selecting for deeper root systems may reduce the need for irrigation.[99]

Content

Nutrition

In a reference amount of 100 grams (3.5 oz), a boiled potato with skin supplies 87 calories and is 77% water, 20% carbohydrates (including 2% dietary fiber in the skin and flesh), 2% protein, and contains negligible fat (table). The protein content is comparable to other starchy vegetable staples, as well as grains.[102]

Boiled potatoes are a rich source (20% or more of the Daily Value, DV) of vitamin B6 (23% DV), and contain a moderate amount of vitamin C (16% DV) and B vitamins, such as thiamine, niacin, and pantothenic acid (10% DV each). Boiled potatoes do not supply significant amounts of dietary minerals (table).

The potato is rarely eaten raw because raw potato starch is poorly digested by humans.[103] Depending on the cultivar and preparation method, potatoes can have a high glycemic index (GI) and so are often excluded from the diets of individuals trying to follow a low-GI diet.[104][102] There is a lack of evidence on the effect of potato consumption on obesity and diabetes.[102]

Potatoes, boiled, cooked in skin, flesh, without salt

Nutritional value per 100 g (3.5 oz)		
Energy	364 kJ (87 kcal)	
Carbohydrates	20.1 g	
Sugars	0.9 g	
Dietary fiber	1.8 g	
Fat	0.1 g	
Protein	1.9 g	
Vitamins and minerals		
Vitamins	**Quantity**	**%DV†**
Thiamine (B_1)	0.11 mg	9%
Riboflavin (B_2)	0.02 mg	2%
Niacin (B_3)	1.44 mg	9%
Pantothenic acid (B_5)	0.52 mg	10%
Vitamin B_6	0.3 mg	18%
Folate (B_9)	10 µg	3%
Vitamin C	13 mg	14%
Minerals	**Quantity**	**%DV†**
Calcium	5 mg	0%
Iron	0.31 mg	2%
Magnesium	22 mg	5%
Manganese	0.14 mg	6%
Phosphorus	44 mg	4%
Potassium	379 mg	13%
Sodium	4 mg	0%
Zinc	0.3 mg	3%
Other constituents	**Quantity**	
Water	77 g	

Link to USDA Database entry (https://fdc.nal.usda.gov/fdc-app.html#/food-details/170438/nutrients)

†Percentages estimated using US recommendations for adults,[100] except for potassium, which is estimated based on expert recommendation from the National Academies.[101]

In the UK, potatoes are not considered by the National Health Service as counting or contributing towards the recommended daily five portions of fruit and vegetables, the 5-A-Day program.[105]

Toxicity

The inedible potato fruit — Tubers exposed to light

Some toxic solanine-rich parts of *S. tuberosum*

Raw potatoes contain toxic glycoalkaloids, of which the most prevalent are solanine and chaconine. Solanine is found in other plants in the same family, Solanaceae, which includes such plants as deadly nightshade (*Atropa belladonna*), henbane (*Hyoscyamus niger*) and tobacco (*Nicotiana* spp.), as well as food plants like tomato. These compounds, which protect the potato plant from its predators, are especially concentrated in the aerial parts of the plant. The tubers are low in these toxins, unless they are exposed to light, which makes them go green.[106][107]

Exposure to light, physical damage, and age increase glycoalkaloid content within the tuber.[108] Different potato varieties contain different levels of glycoalkaloids. The 'Lenape' variety, released in 1967, was withdrawn in 1970 as it contained high levels of glycoalkaloids.[109] Since then, breeders of new varieties test for this, sometimes discarding an otherwise promising cultivar. Breeders try to keep glycoalkaloid levels below 200 mg/kg (0.0032 oz/lb). However, when these commercial varieties turn green, their solanine concentrations can go well above this limit,[110] with higher levels in the potato's skin.[111]

Uses

Culinary

Potato dishes vary around the world. Peruvian cuisine naturally contains the potato as a primary ingredient in many dishes, as around 3,000 varieties of the tuber are grown there.[112] Chuño is a freeze-dried potato product traditionally made by Quechua and Aymara communities of Peru and Bolivia.[113] In the UK, potatoes form part of the traditional dish fish and chips. Roast potatoes are commonly served as part of a Sunday roast dinner and mashed potatoes form a major component of several other traditional dishes, such as shepherd's pie, bubble and squeak, and bangers and mash. New potatoes may be cooked with mint and are often served with butter. In Germany, Northern (Finland, Latvia and especially Scandinavian countries), Eastern Europe (Russia, Belarus and Ukraine) and Poland, newly harvested, early ripening varieties are considered a special delicacy. Boiled whole and served un-peeled with dill, these "new potatoes" are traditionally consumed with Baltic herring. Puddings made from grated potatoes (kugel, kugelis, and potato babka) are popular items of Ashkenazi, Lithuanian, and Belarusian cuisine.[114] Cepelinai, the national dish of Lithuania, are dumplings made from boiled grated potatoes, usually stuffed with minced meat.[115] In Italy, in the Friuli region, potatoes serve to make a type of pasta called gnocchi.[116] Potato is used in northern China where rice is not easily grown, a popular dish being 青椒土豆丝 (*qīng jiāo tǔ dòu sī*), made with green pepper, vinegar and thin slices of potato. In the winter, roadside sellers in northern China sell roasted potatoes.[117]

Pommes frites, also called chips and French fries

Papa rellena

Baked potato with sour cream and chives

German *Bauernfrühstück* ("farmer's breakfast")

Cepelinai

Other uses

Potatoes are sometimes used to brew alcoholic spirits such as vodka, poitín, akvavit, and brännvin.[118][119]

Potatoes are used as fodder for livestock. They may be made into silage which can be stored for some months before use.[120][121]

Potato starch is used in the food industry as a thickener and binder for soups and sauces, in the textile industry as an adhesive, and in the paper industry for the manufacturing of papers and boards.[122][123]

Potatoes are commonly used in plant research. The consistent parenchyma tissue, the clonal nature of the plant and the low metabolic activity make it an ideal model tissue for experiments on wound-response studies and electron transport.[124]

Cultural significance

In mythology

Axomamma, Incan goddess of potatoes

In Inca mythology, a daughter of the earth mother Pachamama, Axomamma, is the goddess of potatoes. She ensured the fertility of the soil and the growth of the tubers.[125] According to Iroquois mythology, the first potatoes grew out of Earth Woman's feet after she died giving birth to her twin sons, Sapling and Flint.[126]

In art

The potato has been an essential crop in the Andes since the pre-Columbian era. The Moche culture from Northern Peru made ceramics from the earth, water, and fire. This pottery was a sacred substance, formed in significant shapes and used to represent important themes. Potatoes are represented anthropomorphically as well as naturally.[127] During the late 19th century, numerous images of potato harvesting appeared in European art, including the works of Willem Witsen and Anton Mauve.[128] Van Gogh's 1885 painting *The Potato Eaters* portrays a family eating potatoes. Van Gogh said he wanted to depict peasants as they really were. He deliberately chose coarse and ugly models, thinking that they would be natural and unspoiled in his finished work.[129] Jean-François Millet's *The Potato Harvest* depicts peasants working in the plains between Barbizon and Chailly. It presents a theme representative of the peasants' struggle for survival. Millet's technique for this work incorporated paste-like pigments thickly applied over a coarsely textured canvas.[130]

Potato ceramic from the Moche culture

The Potato Harvest by Jean-François Millet, 1855 (Walters Art Museum)

The potato harvest by Jules Bastien-Lepage, 1877, National Gallery of Victoria

The Potato Eaters by Van Gogh, 1885 (Van Gogh Museum)

Girl peeling potatoes by Albert Anker, 1886, oil on canvas

In popular culture

Invented in 1949, and marketed and sold commercially by Hasbro in 1952, Mr. Potato Head is an American toy that consists of a plastic potato and attachable plastic parts, such as ears and eyes, to make a face. It was the first toy ever advertised on television.[131][132][133]

In the 2015 fictionalized film, *The Martian*, stranded astronaut and botanist, Mark Watney, cultivates potatoes in the artificial crew habitat using Martian soil fertilized with frozen feces, and produces water from unused rocket fuel.[134]

See also

- Irish potato candy
- List of potato museums
- Loy (spade), a form of early spade used in Ireland for the cultivation of potatoes
- New World crops
- Potato battery
- International Year of the Potato

References

1. Herrero, María Antonieta Andión (2004). *Los indigenismos en la Historia de las Indias de Bartolomé de las Casas* (https://books.google.com/books?id=wwI4nMg5r70C&pg=PA78) (in Spanish). Editorial CSIC - CSIC Press. p. 78. ISBN 978-84-00-08266-6. Archived (https://web.archive.org/web/20240324172324/https://books.google.com/books?id=wwI4nMg5r70C&pg=PA78#v=onepage&q&f=false) from the original on 24 March 2024. Retrieved 2 February 2021.
2. "spud (n.)" (https://www.etymonline.com/word/spud). *Online Etymology Dictionary*. Retrieved 13 May 2018.
3. "jordäpple | SAOB | svenska.se" (https://svenska.se/saob/?sok=jord%C3%A4pple#U_J1_193122) (in Swedish). Retrieved 28 June 2023.
4. Hooshmand, Dana (12 October 2020). " "Earth Apple": The 5 Languages that Use This for "Potato" " (https://discoverdiscomfort.com/earth-apple-potato-languages/). *discoverdiscomfort.com*. Retrieved 27 August 2021.
5. Laws, Christopher (9 February 2015). "A Cultural History of the Potato as Earth Apple" (https://culturedarm.com/cultural-history-potato-earth-apple/). *Culturedarm*. Retrieved 27 August 2021.
6. "Solanum tuberosum: Potato" (https://www.kew.org/plants/potato). Royal Botanic Gardens Kew. Retrieved 5 May 2024.
7. Ewing, E. E.; Struik, P. C. (1992). "Tuber Formation in Potato: Induction, Initiation, and Growth". In Janick, Jules (ed.). *Horticultural Reviews*. pp. 89–198. doi:10.1002/9780470650523.ch3 (https://doi.org/10.1002%2F9780470650523.ch3). ISBN 978-0-471-57339-5.
8. Amador, Virginia; Bou, Jordi; Martínez-García, Jaime; Monte, Elena; Rodríguez-Falcon, Mariana; Russo, Esther; Prat, Salomé (2001). "Regulation of potato tuberization by daylength and gibberellins" (https://web.archive.org/web/20090206110630/http://www.ijdb.ehu.es/abstract.01supp/s37.pdf) (PDF). *International Journal of Developmental Biology* (45): S37–S38. Archived from the original (http://www.ijdb.ehu.es/abstract.01supp/s37.pdf) (PDF) on 6 February 2009. Retrieved 8 January 2009.
9. Plaisted, R. (1982). "Potato". In W. Fehr & H. Hadley (ed.). *Hybridization of Crop Plants*. New York: American Society of Agronomy, Crop Science Society of America. pp. 483–494. ISBN 0-89118-034-6.
10. Olmstead, Richard G., et al. "Phylogeny and provisional classification of the Solanaceae based on chloroplast DNA." Solanaceae IV 1.1 (1999): 1-137. https://www.researchgate.net/profile/Tharindu-Ranasinghe-2/post/Is-there-a-complete-phylogenetic-description-of-the-Solanaceae-family/attachment/59d63db579197b807799a764/AS%3A421051545735172%401477397919618/download/PHYLOGENY
11. Raker, Celeste M.; Spooner, David M. (2002). "Chilean Tetraploid Cultivated Potato, *Solanum tuberosum* is Distinct from the Andean Populations: Microsatellite Data" (https://web.archive.org/web/20090326171403/http://crop.scijournals.org/cgi/reprint/42/5/1451.pdf) (PDF). *Crop Science*. **42**. doi:10.2135/cropsci2002.1451 (https://doi.org/10.2135%2Fcropsci2002.1451). ISSN 0011-183X (https://search.worldcat.org/issn/0011-183X). University of Wisconsin. Archived from the original (http://crop.scijournals.org/cgi/reprint/42/5/1451.pdf) (PDF) on 26 March 2009. Retrieved 16 July 2010.
12. Anabalón Rodríguez, Leonardo; Morales Ulloa, Daniza; Solano Solis, Jaime (July 2007). "Molecular description and similarity relationships among native germplasm potatoes (*Solanum tuberosum* ssp. *tuberosum* L.) using morphological data and AFLP markers" (https://scielo.conicyt.cl/scielo.php?script=sci_arttext&pid=S0717-34582007000300011). *Electronic Journal of Biotechnology*. **10** (3): 436–443. doi:10.2225/vol10-issue3-fulltext-14 (https://doi.org/10.2225%2Fvol10-issue3-fulltext-14). hdl:10925/320 (https://hdl.handle.net/10925%2F320). Retrieved 6 December 2009.
13. Hijmans, R.J.; Spooner, D.M. (2001). "Geographic distribution of wild potato species". *American Journal of Botany*. **88** (11): 2101–12. doi:10.2307/3558435 (https://doi.org/10.2307%2F3558435). JSTOR 3558435 (https://www.jstor.org/stable/3558435). PMID 21669641 (https://pubmed.ncbi.nlm.nih.gov/21669641).
14. Spooner, David M.; McLean, Karen; Ramsay, Gavin; Waugh, Robbie; Bryan, Glenn J. (29 September 2005). "A single domestication for potato based on multilocus amplified fragment length polymorphism genotyping" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1253605). *Proceedings of the National Academy of Sciences*. **102** (41): 14694–14699. Bibcode:2005PNAS..10214694S (https://ui.adsabs.harvard.edu/abs/2005PNAS..10214694S). doi:10.1073/pnas.0507400102 (https://doi.org/10.1073%2Fpnas.0507400102). PMC 1253605 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1253605). PMID 16203994 (https://pubmed.ncbi.nlm.nih.gov/16203994).
15. Office of International Affairs (1989). *Lost Crops of the Incas: Little-Known Plants of the Andes with Promise for Worldwide Cultivation* (http://www.nap.edu/openbook.php?isbn=030904264X&page=92). p. 92. doi:10.17226/1398 (https://doi.org/10.17226%2F1398). ISBN 978-0-309-04264-2.
16. John Michael Francis (2005). *Iberia and the Americas: Culture, Politics, and History : a Multidisciplinary Encyclopedia* (https://books.google.com/books?id=OMNoS-g1h8cC&pg=PA867). ABC-CLIO. p. 867. ISBN 978-1-85109-421-9.
17. Martins-Farias 1976; Moseley 1975
18. Harris, David R.; Hillman, Gordon C. (2014). *Foraging and Farming: The Evolution of Plant Exploitation* (https://books.google.com/books?id=qxghBQAAQBAJ&pg=PA496). Routledge. p. 496. ISBN 978-1-317-59829-9.
19. "Using DNA, scientists hunt for the roots of the modern potato" (https://www.eurekalert.org/news-releases/814749). *EurekAlert!*. Retrieved 23 January 2024.
20. Sauer, Jonathan (2017). *Historical Geography of Crop Plants : a Select Roster*. Boca Raton, FL: CRC Press. p. 320. ISBN 978-0-203-75190-9. OCLC 1014382952 (https://search.worldcat.org/oclc/1014382952). ISBN 9780849389016 ISBN 9781351440622 ISBN 9781351440615 ISBN 9781351440639 ISBN 9780367449872
21. Nunn, Nathan; Qian, Nancy (2011). "The Potato's Contribution to Population and Urbanization: Evidence from a Historical Experiment" (https://web.archive.org/web/20110705043431/http://www.economics.harvard.edu/faculty/nunn/files/Potato_QJE.pdf) (PDF). *Quarterly Journal of Economics*. **126** (2): 593–650. doi:10.1093/qje/qjr009 (https://doi.org/10.1093%2Fqje%2Fqjr009). PMID 22073408 (https://pubmed.ncbi.nlm.nih.gov/22073408). S2CID 17631317 (https://api.semanticscholar.org/CorpusID:17631317). Archived from the original (http://www.economics.harvard.edu/faculty/nunn/files/Potato_QJE.pdf) (PDF) on 5 July 2011. Retrieved 7 July 2012.
22. Nowicki, Marcin; Foolad, Majid R.; Nowakowska, Marzena; Kozik, Elzbieta U.; et al. (17 August 2011). "Potato and tomato late blight caused by *Phytophthora infestans*: An overview of pathology and resistance breeding" (https://doi.org/10.1094%2FPDIS-05-11-0458). *Plant Disease*. **96** (1). American Phytopathological Society: 4–17. doi:10.1094/PDIS-05-11-0458 (https://doi.org/10.1094%2FPDIS-05-11-0458). PMID 30731850 (https://pubmed.ncbi.nlm.nih.gov/30731850).

23. "Cultivated Potato Genebank" (https://cipotato.org/genebankcip/potato-cultivated/). International Potato Center. Retrieved 15 June 2021.
24. Visser, R.G.F.; Bachem, C.W.B.; Boer, J.M.; et al. (2009). "Sequencing the Potato Genome: Outline and First Results to Come from the Elucidation of the Sequence of the World's Third Most Important Food Crop" (https://doi.org/10.1007%2Fs12230-009-9097-8). *American Journal of Potato Research*. **86** (6): 417–29. doi:10.1007/s12230-009-9097-8 (https://doi.org/10.1007%2Fs12230-009-9097-8).
25. Ames, M.; Spooner, D.M. (February 2008). "DNA from herbarium specimens settles a controversy about origins of the European potato" (https://doi.org/10.3732%2Fajb.95.2.252). *American Journal of Botany*. **95** (2): 252–257. doi:10.3732/ajb.95.2.252 (https://doi.org/10.3732%2Fajb.95.2.252). PMID 21632349 (https://pubmed.ncbi.nlm.nih.gov/21632349). S2CID 41052277 (https://api.semanticscholar.org/CorpusID:41052277).
26. Song, J; Bradeen, J.M.; Naess, S.K.; Raasch, J.A.; Wielgus, S.M.; Haberlach, G.T.; Liu, J; Kuang, H; Austin-Phillips, S; Buell, C.R.; Helgeson, J.P.; Jiang, J (2003). "Gene RB cloned from *Solanum bulbocastanum* confers broad spectrum resistance to potato late blight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC170883). *Proceedings of the National Academy of Sciences*. **100** (16): 9128–9133. Bibcode:2003PNAS..100.9128S (https://ui.adsabs.harvard.edu/abs/2003PNAS..100.9128S). doi:10.1073/pnas.1533501100 (https://doi.org/10.1073%2Fpnas.1533501100). PMC 170883 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC170883). PMID 12872003 (https://pubmed.ncbi.nlm.nih.gov/12872003).
27. Paluchowska, Paulina; Sliwka, Jadwiga; Yin, Zhimin (2022). "Late blight resistance genes in potato breeding" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9110483). *Planta*. **255** (6). Springer Science and Business Media LLC: 127. Bibcode:2022Plant.255..127P (https://ui.adsabs.harvard.edu/abs/2022Plant.255..127P). doi:10.1007/s00425-022-03910-6 (https://doi.org/10.1007%2Fs00425-022-03910-6). eISSN 1432-2048 (https://search.worldcat.org/issn/1432-2048). ISSN 0032-0935 (https://search.worldcat.org/issn/0032-0935). PMC 9110483 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9110483). PMID 35576021 (https://pubmed.ncbi.nlm.nih.gov/35576021).
28. Bradshaw, J.; Bryan, G.; Ramsay, G. (2006). "Genetic Resources (Including Wild and Cultivated *Solanum* Species) and Progress in their Utilisation in Potato Breeding". *Potato Research*. **49** (1). Springer Science and Business Media LLC: 49–65. doi:10.1007/s11540-006-9002-5 (https://doi.org/10.1007%2Fs11540-006-9002-5). ISSN 0014-3065 (https://search.worldcat.org/issn/0014-3065). S2CID 30648732 (https://api.semanticscholar.org/CorpusID:30648732).
29. "United Nations: International Day of Potato: 30 May" (https://www.un.org/en/observances/potato-day). United Nations. Retrieved 31 May 2024.
30. Eggers, Ernst-Jan; Burgt, van der; Heusden, van; et al. (6 July 2021). "Neofunctionalisation of the Sli gene leads to self-compatibility and facilitates precision breeding in potato" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8260583). *Nature Communications*. **12** (1): 4141. Bibcode:2021NatCo..12.4141E (https://ui.adsabs.harvard.edu/abs/2021NatCo..12.4141E). doi:10.1038/s41467-021-24267-6 (https://doi.org/10.1038%2Fs41467-021-24267-6). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 8260583 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8260583). PMID 34230471 (https://pubmed.ncbi.nlm.nih.gov/34230471).
31. Hosaka, Kazuyoshi; Hanneman, Robert E. Jr. (1998). "Genetics of self-compatibility in a self-incompatible wild diploid potato species *Solanum chacoense*. 1. Detection of an S locus inhibitor (Sli) gene" (https://link.springer.com/article/10.1023/A:1018353613431). *Euphytica*. **99** (3): 191–197. doi:10.1023/a:1018353613431 (https://doi.org/10.1023%2Fa%3A1018353613431). ISSN 0014-2336 (https://search.worldcat.org/issn/0014-2336). S2CID 40678039 (https://api.semanticscholar.org/CorpusID:40678039).
32. Ma, Ling; Zhang, Chunzhi; Zhang, Bo; et al. (6 July 2021). "A nonS-locus F-box gene breaks self-incompatibility in diploid potatoes" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8260799). *Nature Communications*. **12** (1): 4142. Bibcode:2021NatCo..12.4142M (https://ui.adsabs.harvard.edu/abs/2021NatCo..12.4142M). doi:10.1038/s41467-021-24266-7 (https://doi.org/10.1038%2Fs41467-021-24266-7). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 8260799 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8260799). PMID 34230469 (https://pubmed.ncbi.nlm.nih.gov/34230469).
33. Lindhout, Pim; Meijer, Dennis; Schotte, Theo; Hutten, Ronald C. B.; Visser, Richard G. F.; van Eck, Herman J. (2011). "Towards F1 Hybrid Seed Potato Breeding" (https://doi.org/10.1007%2Fs11540-011-9196-z). *Potato Research*. **54** (4). Springer: 301–312. doi:10.1007/s11540-011-9196-z (https://doi.org/10.1007%2Fs11540-011-9196-z). ISSN 0014-3065 (https://search.worldcat.org/issn/0014-3065). S2CID 39719359 (https://api.semanticscholar.org/CorpusID:39719359).
34. Angmo, Dechen; Sharma, Sat Pal; Kalia, Anu (2023). "Breeding strategies for late blight resistance in potato crop: recent developments". *Molecular Biology Reports*. **50** (9). Springer: 7879–7891. doi:10.1007/s11033-023-08577-0 (https://doi.org/10.1007%2Fs11033-023-08577-0). ISSN 0301-4851 (https://search.worldcat.org/issn/0301-4851). PMID 37526862 (https://pubmed.ncbi.nlm.nih.gov/37526862). S2CID 260349512 (https://api.semanticscholar.org/CorpusID:260349512).
35. Theisen, K (1 January 2007). "History and overview" (https://web.archive.org/web/20080114015939/http://research.cip.cgiar.org/confluence/display/wpa/Peru). *World Potato Atlas: Peru*. International Potato Center. Archived from the original (http://research.cip.cgiar.org/confluence/display/wpa/Peru) on 14 January 2008. Retrieved 10 September 2008.
36. "*Solanum tuberosum* L." (http://powo.science.kew.org/taxon/urn:lsid:ipni.org:names:821337-1#synonyms) *Plants of the World Online*. Royal Botanic Gardens, Kew. 2017. Retrieved 7 September 2020.
37. "Europotato.org" (https://web.archive.org/web/20091128021457/http://www.europotato.org/menu.php). Europotato.org. Archived from the original (http://www.europotato.org/menu.php) on 28 November 2009. Retrieved 16 July 2010.
38. Potato Council. "Potato Varieties" (https://web.archive.org/web/20090908212045/http://www.britishpotatoes.co.uk/potato-varieties). Agriculture & Horticulture Development Board. Archived from the original (http://www.britishpotatoes.co.uk/potato-varieties/) on 8 September 2009. Retrieved 13 September 2009.
39. "Potato Primer" (https://web.archive.org/web/20081217030233/http://www.cooksillustrated.com/images/document/howto/JF07_PotatoPrimer.pdf) (PDF). Cooks Illustrated. Archived from the original (http://www.cooksillustrated.com/images/document/howto/JF07_PotatoPrimer.pdf) (PDF) on 17 December 2008. Retrieved 8 December 2008.
40. Agricultural Marketing Service. "Potatoes for Chipping Grades and Standards" (https://www.ams.usda.gov/grades-standards/potatoes-chipping-grades-and-standards). Retrieved 27 August 2018.
41. "Creamer Potato" (http://www.recipetips.com/glossary-term/t--35863/creamer-potato.asp). recipetips.com. Retrieved 18 July 2008.
42. "What is a new potato? New guidelines issued" (https://www.bbc.co.uk/news/uk-scotland-glasgow-west-23667664). *BBC News*. 12 August 2013. Retrieved 13 June 2021.
43. "A look back at a Royal history" (http://news.bbc.co.uk/local/jersey/hi/people_and_places/nature/newsid_8478000/8478833.stm). 25 January 2010. Retrieved 13 June 2021.
44. "So many varieties, so many choices" (http://wisconsinpotatoes.com/growing/varieties/). Wisconsin Potato and Vegetable Growers Association. 2017.
45. Hirsch, C.N.; Hirsch, C.D.; Felcher, K; Coombs, J; Zarka, D; Van Deynze, A; De Jong, W; Veilleux, R.E.; Jansky, S; Bethke, P; Douches, D.S.; Buell, C.R. (2013). "Retrospective View of North American Potato (*Solanum tuberosum* L.) Breeding in the 20th and 21st Centuries" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3689798). *G3: Genes, Genomes, Genetics*. **3** (6): 1003–13. doi:10.1534/g3.113.005595 (https://doi.org/10.1534%2Fg3.113.005595). PMC 3689798 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3689798). PMID 23589519 (https://pubmed.ncbi.nlm.nih.gov/23589519).
46. Jemison, John M. Jr.; Sexton, Peter; Camire, Mary Ellen (2008). "Factors Influencing Consumer Preference of Fresh Potato Varieties in Maine". *American Journal of Potato Research*. **85** (2): 140. doi:10.1007/s12230-008-9017-3 (https://doi.org/10.1007%2Fs12230-008-9017-3). S2CID 34297429 (https://api.semanticscholar.org/CorpusID:34297429).
47. Mattoo, A.K.; Shukla, V; Fatima, T; Handa, A.K.; Yachha, S.K. (2010). "Genetic Engineering to Enhance Crop-Based Phytonutrients (Nutraceuticals) to Alleviate Diet-Related Diseases". *Bio-Farms for Nutraceuticals*. Advances in Experimental Medicine and Biology. Vol. 698. pp. 122–43. doi:10.1007/978-1-4419-7347-4_10 (https://doi.org/10.1007%2F978-1-4419-7347-4_10). ISBN 978-1-4419-7346-7. PMID 21520708 (https://pubmed.ncbi.nlm.nih.gov/21520708).
48. "Genetically Engineered Organisms Public Issues Education Project/Am I eating GE potatoes?" (https://web.archive.org/web/20090103033654/http://www.geo-pie.cornell.edu/crops/potato.html). Cornell University. Archived from the original (http://www.geo-pie.cornell.edu/crops/potato.html) on 3 January 2009. Retrieved 16 December 2008.
49. "GMO compass database" (https://web.archive.org/web/20141009210148/http://www.gmo-compass.org/eng/gmo/popups/55.potato_eh92_527_1.html). Archived from the original (http://www.gmo-compass.org/eng/gmo/popups/55.potato_eh92_527_1.html) on 9 October 2014. Retrieved 6 October 2014.
50. "GM potatoes: BASF at work" (https://web.archive.org/web/20100531073525/http://www.gmo-compass.org/eng/news/492.docu.html). 31 May 2010. Archived from the original (http://www.gmo-compass.org/eng/news/492.docu.html) on 31 May 2010.
51. Oh, Soohyun; Choi, Doil (2022). "Receptor-mediated nonhost resistance in plants" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9528085). Review. *Essays in Biochemistry*. **66** (5). Portland Press Limited (Biochemical Society): 435–445. doi:10.1042/EBC20210080 (https://doi.org/10.1042%2FEBC20210080). PMC 9528085 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9528085). PMID 35388900 (https://pubmed.ncbi.nlm.nih.gov/35388900). S2CID 247999992 (https://api.semanticscholar.org/CorpusID:247999992).
52. "Research in Germany: Business BASF applies for approval for another biotech potato" (https://web.archive.org/web/20130602111343/http://www.research-in-germany.de/84190/2011-11-17-business-basf-applies-for-approval-for-another-biotech-potato.html). 2 June 2013. Archived from the original (http://www.research-in-germany.de/84190/2011-11-17-business-basf-applies-for-approval-for-another-biotech-potato.html) on 2 June 2013.
53. Burger, Ludwig (10 November 2015). "BASF applies for EU approval for Fortuna GM potato" (https://web.archive.org/web/20151110105537/http://www.reuters.com/article/2011/10/31/us-basf-idUSTRE79U41Q20111031). *Reuters*. Archived from the original (http://www.reuters.com/article/2011/10/31/us-basf-idUSTRE79U41Q20111031) on 10 November 2015.
54. BASF stops GM crop development in Europe (http://www.dw.de/basf-stops-gm-crop-development-in-europe/a-15671900), *Deutsche Welle*, 17 January 2012
55. Kanter, James (16 January 2012). "BASF to Stop Selling Genetically Modified Products in Europe" (https://archive.today/20240123213954/https://www.nytimes.com/2012/01/17/business/global/17iht-gmo17.html). *The New York Times*. Archived from the original (https://www.nytimes.com/2012/01/17/business/global/17iht-gmo17.html) on 23 January 2024. Retrieved 23 January 2024.

56. Pollack, Andrew (7 November 2014). "U.S.D.A. Approves Modified Potato. Next Up: French Fry Fans" (https://archive.today/20141112021024/http://www.nytimes.com/2014/11/08/business/genetically-modified-potato-from-simplot-approved-by-usda.html?_r=0). *The New York Times*. Archived from the original (https://www.nytimes.com/2014/11/08/business/genetically-modified-potato-from-simplot-approved-by-usda.html) on 12 November 2014.
57. "Consumer acceptance of genetically modified potatoes" (https://web.archive.org/web/20121101142414/http://www.agbioforum.org/v7n12/v7n12a13-mccluskey.pdf) (PDF). *American Journal of Potato Research*. 2002. cited through Bnet. Archived from the original (http://www.agbioforum.org/v7n12/v7n12a13-mccluskey.pdf) (PDF) on 1 November 2012. Retrieved 19 February 2012.
58. Rosenthal, Elisabeth (24 July 2007). "A genetically modified potato, not for eating, is stirring some opposition in Europe" (https://www.nytimes.com/2007/07/24/business/worldbusiness/24spuds.html). *The New York Times*. Retrieved 15 November 2008.
59. United States Potato Board. "Seed Potatoes" (https://web.archive.org/web/20150825053643/http://www.potatoesusa.com/potato-products/seed-potatoes). Archived from the original (http://www.potatoesusa.com/potato-products/seed-potatoes) on 25 August 2015. Retrieved 6 October 2014.
60. "Seed & Ware Potatoes – Nuclear Stock" (https://www.sasa.gov.uk/seed-ware-potatoes/nuclear-stock). *SASA*. Archived (https://web.archive.org/web/20150910164423/http://www.sasa.gov.uk/seed-ware-potatoes/nuclear-stock) from the original on 10 September 2015. Retrieved 12 June 2024.
61. "Potatoes Home Garden" (https://sfyl.ifas.ufl.edu/lawn-and-garden/potatoes-home-garden/). *sfyl.ifas.ufl.edu*. UF/IFAS Extension. Retrieved 14 August 2019.
62. Jefferies, R. A.; Lawson, H. M. (1991). "A key for the stages of development of potato (*Solanum tuberosum*)". *Annals of Applied Biology*. **119** (2): 387–399. doi:10.1111/j.1744-7348.1991.tb04879.x (https://doi.org/10.1111%2Fj.1744-7348.1991.tb04879.x). ISSN 0003-4746 (https://search.worldcat.org/issn/0003-4746).
63. "Growing Potatoes in the Home Garden" (https://web.archive.org/web/20110516020007/http://suffolk-lamp.cit.cornell.edu/assets/Horticulture-Leaflets/Growing-Potatoes-in-the-Home-Garden.pdf) (PDF). Cornell University Extension Service. Archived from the original (http://suffolk-lamp.cit.cornell.edu/assets/Horticulture-Leaflets/Growing-Potatoes-in-the-Home-Garden.pdf) (PDF) on 16 May 2011. Retrieved 27 June 2010.
64. "Potato Production USAID-Inma -- POTATO PRODUCTION: PLANTING THROUGH HARVEST" (https://pdf.usaid.gov/pdf_docs/pnadz110.pdf) (PDF). USAID. pp. 2–21. Retrieved 4 June 2024.
65. "How to grow potatoes: Planting" (https://www.rhs.org.uk/vegetables/potatoes/grow-your-own). Royal Horticultural Society. Retrieved 4 June 2024.
66. "Potatoes" (https://www.nsalg.org.uk/crop/potatoes/). The National Allotment Society. Retrieved 5 May 2024.
67. "Organic Management of Late Blight of Potato and Tomato (*Phytophthora infestans*)" (https://web.archive.org/web/20150702002959/http://www.extension.org/pages/18361/organic-management-of-late-blight-of-potato-and-tomato-phytophthora-infestans). Michigan State University. Archived from the original (http://www.extension.org/pages/18361/organic-management-of-late-blight-of-potato-and-tomato-phytophthora-infestans) on 2 July 2015. Retrieved 6 January 2012.
68. "Potato, Identifying Diseases" (https://ag.umass.edu/vegetable/fact-sheets/potato-identifying-diseases). University of Massachusetts Amherst. 15 November 2016. Retrieved 5 May 2024.
69. "Potato Disease Identification" (https://potatoes.ahdb.org.uk/knowledge-library/potato-disease-identification). Agriculture and Horticulture Development Board. Retrieved 5 May 2024.
70. Alyokhin, A. (2009). "Colorado potato beetle management on potatoes: current challenges and future prospects" (http://www.potatobeetle.org/Alyokhin_CPB_Review_reprint.pdf) (PDF). In Tennant, P.; Benkeblia, N. (eds.). *Potato II. Fruit, Vegetable and Cereal Science and Biotechnology 3 (Special Issue 1)*. pp. 10–19.
71. "Potato Cyst Nematode" (https://agriculture.vic.gov.au/agriculture/pests-diseases-and-weeds/pest-insects-and-mites/potato-cyst-nematode). *Agriculture Victoria*. Archived (https://web.archive.org/web/20191202063605/http://agriculture.vic.gov.au/agriculture/pests-diseases-and-weeds/pest-insects-and-mites/potato-cyst-nematode) from the original on 2 December 2019. Retrieved 11 December 2019.
72. Johnson, Ciaran Miceal; Auat Cheein, Fernando (2023). "Machinery for potato harvesting: a state-of-the-art review" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10239890). *Frontiers in Plant Science*. **14**. doi:10.3389/fpls.2023.1156734 (https://doi.org/10.3389%2Ffpls.2023.1156734). ISSN 1664-462X (https://search.worldcat.org/issn/1664-462X). PMC 10239890 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10239890). PMID 37284722 (https://pubmed.ncbi.nlm.nih.gov/37284722).
73. Kleinkopf, G.E.; Olsen, N. (2003). "Storage Management". In J.C. Stark; S.L. Love (eds.). *Potato Production Systems*. University of Idaho Agricultural Communications. pp. 363–381.
74. Potato storage, value Preservation: Kohli, Pawanexh (2009). "Potato storage and value Preservation: The Basics" (https://web.archive.org/web/20200806192307/http://www.crosstree.info/Documents/POTATO_STORAGE.pdf) (PDF). CrossTree techno-visors. Archived from the original (http://crosstree.info/Documents/POTATO_STORAGE.pdf) (PDF) on 6 August 2020. Retrieved 12 July 2009.
75. "Can eating burnt foods cause cancer?" (https://www.cancerresearchuk.org/about-cancer/causes-of-cancer/cancer-myths/can-eating-burnt-foods-cause-cancer). Cancer Research UK. 15 October 2021.
76. Epp, Melanie (12 April 2021). "The Worry with CIPC" (https://european-seed.com/2021/04/the-worry-with-cipc/). *EuropeanSeed*. Retrieved 12 June 2021.
77. "Potato production in 2021 Region/World/Production Quantity/Crops from pick lists" (https://www.fao.org/faostat/en/#data/QCL). UN Food and Agriculture Organization, Statistics Division. 2023. Retrieved 18 April 2023.
78. *World Food and Agriculture – Statistical Yearbook 2021* (https://www.fao.org/documents/card/en/c/cb4477en/). Geneva: FAOSTAT. 2021. doi:10.4060/cb4477en (https://doi.org/10.4060%2Fcb4477en). ISBN 978-92-5-134332-6. S2CID 240163091 (https://api.semanticscholar.org/CorpusID:240163091). Retrieved 13 December 2021.
79. "FAOSTAT: Production-Crops, 2010 data" (https://web.archive.org/web/20130114151638/http://faostat.fao.org/site/567/DesktopDefault.aspx?PageID=567). Food and Agriculture Organization of the United Nations. 2011. Archived from the original (http://faostat.fao.org/site/567/DesktopDefault.aspx?PageID=567#ancor) on 14 January 2013.
80. Sarah Sinton (2011). "There's yet more gold in them thar "hills"!" (http://maxa.maf.govt.nz/sff/about-projects/search/05-157/grower-article.htm). Grower Magazine, The Government of New Zealand.
81. "Phosphate and potatoes" (https://web.archive.org/web/20120301060943/http://www.ballance.co.nz/technical+expertise/horticulture/phosphate+and+potatoes). Ballance. 2009. Archived from the original (http://www.ballance.co.nz/technical+expertise/horticulture/phosphate+and+potatoes) on 1 March 2012. Retrieved 19 February 2012.
82. "International Year of the Potato: 2008, Asia and Oceania" (https://web.archive.org/web/20120622203024/http://www.potato2008.org/en/world/asia.html). Potato World. 2008. Archived from the original (http://www.potato2008.org/en/world/asia.html) on 22 June 2012. Retrieved 19 February 2012.
83. *Workshop to Commemorate the International Year of the Potato* (http://www.fao.org/docrep/010/i0200e/I0200E10.htm). The Food and Agriculture Organization of the United Nations. 2008.
84. Foley, Ramankutty; et al. (12 October 2011). "Solutions for a cultivated planet" (http://www.escholarship.org/uc/item/6xw5g085). *Nature*. **478** (7369): 337–42. Bibcode:2011Natur.478..337F (https://ui.adsabs.harvard.edu/abs/2011Natur.478..337F). doi:10.1038/nature10452 (https://doi.org/10.1038%2Fnature10452). PMID 21993620 (https://pubmed.ncbi.nlm.nih.gov/21993620). S2CID 4346486 (https://api.semanticscholar.org/CorpusID:4346486).
85. Ensminger, Audrey; Ensminger, M.E.; Konlande, James E. (1994). *Foods & Nutrition Encyclopedia* (https://books.google.com/books?id=XMA9gYIj-C4C&pg=PA1104). CTC Press. p. 1104. ISBN 978-0-8493-8981-8.
86. Haverkort, A.J.; Verhagen, A. (October 2008). "Climate Change and Its Repercussions for the Potato Supply Chain". *Potato Research*. **51** (3–4): 223–237. doi:10.1007/s11540-008-9107-0 (https://doi.org/10.1007%2Fs11540-008-9107-0). S2CID 22794078 (https://api.semanticscholar.org/CorpusID:22794078).
87. Zhao, Chuang; Liu, Bing; Piao, Shilong; Wang, Xuhui; Lobell, David B.; Huang, Yao; et al. (15 August 2017). "Temperature increase reduces global yields of major crops in four independent estimates" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5584412). *Proceedings of the National Academy of Sciences of the United States of America*. **114** (35): 9326–9331. Bibcode:2017PNAS..114.9326Z (https://ui.adsabs.harvard.edu/abs/2017PNAS..114.9326Z). doi:10.1073/pnas.1701762114 (https://doi.org/10.1073%2Fpnas.1701762114). PMC 5584412 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5584412). PMID 28811375 (https://pubmed.ncbi.nlm.nih.gov/28811375).
88. "Potato" (http://cipotato.org/potato). CIP. Retrieved 7 November 2012.
89. Luck, J.; Spackman, M.; Freeman, A.; Tre˛bicki, P.; Griffiths, W.; Finlay, K.; Chakraborty, S. (2011). "Climate change and diseases of food crops". *Plant Pathology*. **60** (1): 113–121. doi:10.1111/j.1365-3059.2010.02414.x (https://doi.org/10.1111%2Fj.1365-3059.2010.02414.x). ISSN 0032-0862 (https://search.worldcat.org/issn/0032-0862).
90. "Climate change and potatoes: The risks, impacts and opportunities for UK potato production" (https://web.archive.org/web/20120910123323/http://www.potato.org.uk/sites/default/files/%5Bcurrent-page%3Aarg%3A%3F%5D/CC%20impacts%20potatoes_Final_20Sept2011.pdf) (PDF). Cranfield Water Science Institute. Archived from the original (http://www.potato.org.uk/sites/default/files/%5Bcurrent-page%3Aarg%3A%3F%5D/CC%20impacts%20potatoes_Final_20Sept2011.pdf) (PDF) on 10 September 2012. Retrieved 7 November 2012.
91. "Crop Water Information: Potato" (http://www.fao.org/nr/water/cropinfo_potato.html). FAO Water Development and Management Unit. Retrieved 7 November 2012.
92. Daccache, A.; Keay, C.; Jones, R. J. A.; Weatherhead, E. K.; Stalham, M. A.; Knox, J. W. (2012). "Climate change and land suitability for potato production in England and Wales: impacts and adaptation". *The Journal of Agricultural Science*. **150** (2): 161–177. doi:10.1017/S0021859611000839 (https://doi.org/10.1017%2FS0021859611000839). hdl:1826/8188 (https://hdl.handle.net/1826%2F8188). ISSN 0021-8596 (https://search.worldcat.org/issn/0021-8596).
93. Hijmans, Robert J. (2003). "The Effect of Climate Change on Global Potato Production". *American Journal of Potato Research*. **80** (4): 271–280. doi:10.1007/bf02855363 (https://doi.org/10.1007%2Fbf02855363). S2CID 3355406 (https://api.semanticscholar.org/CorpusID:3355406).
94. Levy, D.; Veilleux, R.E. (2007). "Adaptation of Potato to High Temperatures and Salinity A Review". *American Journal of Potato Research*. **84** (6): 487–506. doi:10.1007/bf02987885 (https://doi.org/10.1007%2Fbf02987885). S2CID 602971 (https://api.semanticscholar.org/CorpusID:602971).

95. Pandey, S.K. "Potato Research Priorities in Asia and the Pacific Region" (http://www.fao.org/docrep/010/i0200e/I0200E08.htm). FAO. Retrieved 7 November 2012.
96. Czajkowski, R. "Why is Dickeya spp. (syn. Erwinia chrysanthemi) taking over? The ecology of a blackleg pathogen" (http://www.knpv.org/db/upload/documents/Pests%20and%20climate%20change/van_der_Wolf_KNPV3_dec_2008.pdf) (PDF). Retrieved 7 November 2012.
97. Forbes, G.A. "Implications for a warmer, wetter world on the late blight pathogen: How CIP efforts can reduce risk for low-input potato farmers" (https://web.archive.org/web/20110105075912/http://www.icrisat.org/journal/SpecialProject/sp4.pdf) (PDF). CIP. Archived from the original (http://www.icrisat.org/journal/SpecialProject/sp4.pdf) (PDF) on 5 January 2011. Retrieved 7 November 2012.
98. "Information highlights from World Potato Congress, Kunming, China, April 2004" (https://web.archive.org/web/20130410010504/http://peracto.com.au/publications/world-potato-congress.pdf) (PDF). World Potato Congress. Archived from the original (http://www.peracto.com.au/publications/world-potato-congress.pdf) (PDF) on 10 April 2013. Retrieved 7 November 2012.
99. "Potato and water resources" (https://web.archive.org/web/20120620192756/http://www.potato2008.org/en/potato/water.html). FAO. Archived from the original (http://www.potato2008.org/en/potato/water.html) on 20 June 2012. Retrieved 7 November 2012.
100. United States Food and Drug Administration (2024). "Daily Value on the Nutrition and Supplement Facts Labels" (https://www.fda.gov/food/nutrition-facts-label/daily-value-nutrition-and-supplement-facts-labels). *FDA*. Archived (https://web.archive.org/web/20240327175201/https://www.fda.gov/food/nutrition-facts-label/daily-value-nutrition-and-supplement-facts-labels) from the original on 27 March 2024. Retrieved 28 March 2024.
101. National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Food and Nutrition Board; Committee to Review the Dietary Reference Intakes for Sodium and Potassium (2019). Oria, Maria; Harrison, Meghan; Stallings, Virginia A. (eds.). *Dietary Reference Intakes for Sodium and Potassium* (http://www.ncbi.nlm.nih.gov/books/NBK538102/). The National Academies Collection: Reports funded by National Institutes of Health. Washington, DC: National Academies Press (US). ISBN 978-0-309-48834-1. PMID 30844154 (https://pubmed.ncbi.nlm.nih.gov/30844154). Archived (https://web.archive.org/web/20240509063633/https://www.ncbi.nlm.nih.gov/books/NBK538102/) from the original on 9 May 2024. Retrieved 21 June 2024.
102. Beals, Katherine A. (2019). "Potatoes, Nutrition and Health" (https://doi.org/10.1007%2Fs12230-018-09705-4). *American Journal of Potato Research*. **96** (2): 102–110. doi:10.1007/s12230-018-09705-4 (https://doi.org/10.1007%2Fs12230-018-09705-4).
103. Beazell, JM; Schmidt, CR; Ivy, AC (January 1939). "On the Digestibility of Raw Potato Starch in Man". *The Journal of Nutrition*. **17** (1): 77–83. doi:10.1093/jn/17.1.77 (https://doi.org/10.1093%2Fjn%2F17.1.77).
104. Fernandes, Glen; Velangi, Amogh; Wolever, Thomas M.S. (2005). "Glycemic index of potatoes commonly consumed in North America". *Journal of the American Dietetic Association*. **105** (4): 557–562. doi:10.1016/j.jada.2005.01.003 (https://doi.org/10.1016%2Fj.jada.2005.01.003). PMID 15800557 (https://pubmed.ncbi.nlm.nih.gov/15800557).
105. "5 A Day: what counts?" (https://www.nhs.uk/live-well/eat-well/5-a-day/5-a-day-what-counts/). *nhs.uk*. 23 February 2022. Retrieved 23 January 2024.
106. "Tomato-like Fruit on Potato Plants" (https://web.archive.org/web/20040716065133/http://www.ipm.iastate.edu/ipm/hortnews/2004/7-2-2004/tomatopotato.html). Iowa State University. Archived from the original (http://www.ipm.iastate.edu/ipm/hortnews/2004/7-2-2004/tomatopotato.html) on 16 July 2004. Retrieved 8 January 2009.
107. Friedman, Mendel; McDonald, Gary M.; Filadelfi-Keszi, MaryAnn (1997). "Potato Glycoalkaloids: Chemistry, Analysis, Safety, and Plant Physiology". *Critical Reviews in Plant Sciences*. **16** (1): 55–132. Bibcode:1997CRvPS..16...55F (https://ui.adsabs.harvard.edu/abs/1997CRvPS..16...55F). doi:10.1080/07352689709701946 (https://doi.org/10.1080%2F07352689709701946). ISSN 0735-2689 (https://search.worldcat.org/issn/0735-2689).
108. "Greening of potatoes" (https://web.archive.org/web/20111125205141/http://www.csiro.au/resources/green-potatoes). Food Science Australia. 2005. Archived from the original (http://www.csiro.au/resources/green-potatoes) on 25 November 2011. Retrieved 15 November 2008.
109. Koerth-Baker, Marggie (25 March 2013). "The case of the poison potato" (https://web.archive.org/web/20151108070908/http://boingboing.net/2013/03/25/the-case-of-the-poison-potato.html). boingboing.net. Archived from the original (https://boingboing.net/2013/03/25/the-case-of-the-poison-potato.html) on 8 November 2015. Retrieved 8 November 2015.
110. Friedman, Mendel; Roitman, James N.; Kozukue, Nobuyuki (7 May 2003). "Glycoalkaloid and calystegine contents of eight potato cultivars" (https://pubmed.ncbi.nlm.nih.gov/12720378/). *Journal of Agricultural and Food Chemistry*. **51** (10): 2964–2973. doi:10.1021/jf021146f (https://doi.org/10.1021%2Fjf021146f). ISSN 0021-8561 (https://search.worldcat.org/issn/0021-8561). PMID 12720378 (https://pubmed.ncbi.nlm.nih.gov/12720378).
111. Shaw, Ian (2005). *Is it Safe to Eat?: Enjoy Eating and Minimize Food Risks* (https://books.google.com/books?id=XlfVw2QmdvIC&pg=PA129). Berlin: Springer Science & Business Media. p. 129. ISBN 978-3-540-21286-7.
112. Hayes, Monte (24 June 2007). "*Peru Celebrates Potato Diversity*" (https://www.washingtonpost.com/wp-dyn/content/article/2007/06/24/AR2007062400727.html). *The Washington Post*. Retrieved 16 July 2010.
113. Timothy Johns: With bitter Herbs They Shall Eat it : Chemical ecology and the origins of human diet and medicine, The University of Arizona Press, Tucson 1990, ISBN 0-8165-1023-7, pp. 82–84
114. von Bremzen, Anya; Welchman, John (1990). *Please to the Table: The Russian Cookbook* (https://archive.org/details/pleasetotablethe00vonb/page/319). New York: Workman Publishing. pp. 319–20 (https://archive.org/details/pleasetotablethe00vonb/page/319). ISBN 978-0-89480-845-6.
115. "D.E.L.A.C." (https://web.archive.org/web/20160305044428/http://www.delac.eu/stories/40?back=) *delac.eu*. Archived from the original (http://www.delac.eu/stories/40?back=) on 5 March 2016. Retrieved 25 January 2015.
116. Roden, Claudia (1990). *The Food of Italy*. London: Arrow Books. p. 72. ISBN 978-0-09-976220-1.
117. Solomon, Charmaine (1996). *Charmaine Solomon's Encyclopedia of Asian Food*. Melbourne: William Heinemann Australia. p. 293. ISBN 978-0-85561-688-5.
118. Ermochkine, Nicholas and Iglikowski, Peter (2003). *40 degrees east : an anatomy of vodka*, Nova Publishers, p. 65, ISBN 1-59033-594-5.
119. Brännvinsbränning (https://runeberg.org/nfbd/0233.html) Archived (https://web.archive.org/web/20130621032823/http://runeberg.org/nfbd/0233.html) 21 June 2013 at the Wayback Machine in *Nordisk familjebok*, volume 4 (1905)
120. Halliday, Les; et al. (2015). "Ensiling Potatoes" (http://www.gov.pe.ca/photos/original/af_fact_ensipot.pdf) (PDF). *Prince Edward Island Agriculture and Fisheries*. Retrieved 27 January 2018.
121. Schroeder, Ken (October 2012). "Feeding Cull Potatoes to Dairy and Beef Cattle" (https://fyi.extension.wisc.edu/wbic/files/2012/10/Feeding-Cull-Potatoes-to-Dairy-and-Beef-Cattle-10-24-12.pdf) (PDF). *University of Wisconsin Extension*. Retrieved 4 June 2024.
122. Grant M. Campbell; Colin Webb; Stephen L. McKee (1997). *Cereals: Novel Uses and Processes* (https://books.google.com/books?id=W4o7IUKSxyQC&pg=PA22). Springer. p. 22. ISBN 978-0-306-45583-4.
123. Jai Gopal; S.M. Paul Khurana (2006). *Handbook of Potato Production, Improvement, and Postharvest* (https://books.google.com/books?id=hxy8pkP26NEC&pg=PA544). Haworth Press. p. 544. ISBN 978-1-56022-272-9.
124. Espinoza, N. O.; Estrada, R.; Silva-Rodriguez, D.; Tovar, P.; Lizarraga, R.; Dodds, J. H. (1986). "The Potato: A Model Crop Plant for Tissue Culture". *Outlook on Agriculture*. **15** (1): 21–26. Bibcode:1986OutAg..15...21E (https://ui.adsabs.harvard.edu/abs/1986OutAg..15...21E). doi:10.1177/003072708601500104 (https://doi.org/10.1177%2F003072708601500104). ISSN 0030-7270 (https://search.worldcat.org/issn/0030-7270).
125. Thurner, Mark; Pimentel, Juan (2021). *New World Objects of Knowledge* (https://library.oapen.org/bitstream/handle/20.500.12657/55756/1/9781908857828.pdf#page=267) (PDF). London: Institute of Latin American Studies, School of Advanced Study, University of London. p. 248. ISBN 978-1-908857-82-8. OCLC 1242739583 (https://search.worldcat.org/oclc/1242739583).
126. Converse, Harriet Maxwell (Ya-ie-wa-no); Parker, Arthur Caswell (Ga-wa-so-wa-neh) (15 December 1908). "Myths and Legends of the New York State Iroquois" (https://archive.org/details/cu31924055492973). *Education Department Bulletin*. University of the State of New York: 31–41 "Creation: Ata-en-sic, the Sky Woman" and "The Sun, Moon and Stars". Retrieved 1 June 2024.
127. Berrin, Katherine & Larco Museum. The Spirit of Ancient Peru: Treasures from the Museo Arqueológico Rafael Larco Herrera. New York:Thames and Hudson, 1997.
128. Steven Adams; Anna Gruetzner Robins (2000). *Gendering Landscape Art* (https://books.google.com/books?id=dY7xwrA-ibQC&pg=PA67). University of Manchester. p. 67. ISBN 978-0-7190-5628-4.
129. van Tilborgh, Louis (2009). "The Potato Eaters by Vincent van Gogh" (http://www.vggallery.com/visitors/004.htm). *The Vincent van Gogh Gallery*. Retrieved 11 September 2009.
130. Johnston, W.R., Nineteenth Century Art: From Romanticism to Art Nouveau, The Walters Art Gallery, p.56, ISBN 1857592433
131. "Mr Potato Head" (https://collections.vam.ac.uk/item/O1120879/mr-potato-head-construction-toy-lerner-george/). *Museum of Childhood*. V&A Museum of Childhood. Retrieved 4 June 2024.
132. "About Mr. Potato Head" (https://web.archive.org/web/20080925085901/http://www.hasbro.com/default.cfm?page=ci_history_mph). Archived from the original (http://www.hasbro.com/default.cfm?page=ci_history_mph) on 25 September 2008. Retrieved 28 August 2008.
133. Walsh, Tim (2005). *Timeless Toys: Classic Toys and the Playmakers Who Made Them*. Andrews McMeel Publishing. ISBN 0-7407-5571-4.
134. "Could we grow potatoes on Mars?" (https://warwick.ac.uk/newsandevents/knowledgecentre/science/physics-astrophysics/growing_potatoes_on_mars/). Knowledge Centre, University of Warwick. 18 August 2020. Retrieved 4 June 2024.

Further reading

- *Atlas of Wild Potatoes* (https://cgspace.cgiar.org/bitstream/handle/10568/105275/Atlas_of_wild_potatoes_826.pdf) (2002), Systematic and Ecogeographic Studies on Crop Genepools 10, International Plant Genetic Resources Institute

(IPGRI), ISBN 9789290435181
- *Economist*. "Llamas and mash", *The Economist* 28 February 2008 (http://www.economist.com/world/la/displaystory.cfm?story_id=10766599)
- *Economist*. "The potato: Spud we like", (leader) *The Economist* 28 February 2008 (http://www.economist.com/opinion/displaystory.cfm?story_id=10766030)
- Bohl, William H.; Johnson, Steven B., eds. (2010). *Commercial Potato Production in North America: The Potato Association of America Handbook* (https://web.archive.org/web/20120816144218/http://potatoassociation.org/documents/A_ProductionHandbook_Final_000.pdf) (PDF). Second Revision of American Potato Journal Supplement Volume 57 and USDA Handbook 267. The Potato Association of America. Archived from the original (http://potatoassociation.org/documents/A_ProductionHandbook_Final_000.pdf) (PDF) on 16 August 2012.
- Boomgaard, Peter (2003). "In the Shadow of Rice: Roots and Tubers in Indonesian History, 1500–1950". *Agricultural History*. **77** (4): 582–610. doi:10.1525/ah.2003.77.4.582 (https://doi.org/10.1525%2Fah.2003.77.4.582). JSTOR 3744936 (https://www.jstor.org/stable/3744936).
- Gauldie, Enid (1981). The Scottish Miller 1700–1900. Pub. John Donald. ISBN 0-85976-067-7.
- Hawkes, J.G. (1990). *The Potato: Evolution, Biodiversity & Genetic Resources*, Smithsonian Institution Press, Washington, DC
- Lang, James (1975). *Notes of a Potato Watcher* (https://archive.org/details/notesofpotatowat0000lang). Texas A&M University Agriculture series. ISBN 978-1-58544-138-9.
- Langer, William L (1975). "American Foods and Europe's Population Growth 1750–1850". *Journal of Social History*. **8** (2): 51–66. doi:10.1353/jsh/8.2.51 (https://doi.org/10.1353%2Fjsh%2F8.2.51). JSTOR 3786266 (https://www.jstor.org/stable/3786266).
- McNeill, William H. "How the Potato Changed the World's History." *Social Research* (1999) 66#1 pp. 67–83. ISSN 0037-783X (https://www.worldcat.org/search?fq=x0:jrnl&q=n2:0037-783X) Fulltext: Ebsco, by a leading historian
- McNeill William H (1948). "The Introduction of the Potato into Ireland". *Journal of Modern History*. **21** (3): 218–21. doi:10.1086/237272 (https://doi.org/10.1086%2F237272). JSTOR 1876068 (https://www.jstor.org/stable/1876068). S2CID 145099646 (https://api.semanticscholar.org/CorpusID:145099646).
- Ó Gráda, Cormac. *Black '47 and Beyond: The Great Irish Famine in History, Economy, and Memory.* (1999). 272 pp.
- Ó Gráda, Cormac, Richard Paping, and Eric Vanhaute, eds. *When the Potato Failed: Causes and Effects of the Last European Subsistence Crisis, 1845–1850.* (2007). 342 pp. ISBN 978-2-503-51985-2. 15 essays by scholars looking at Ireland and all of Europe
- Reader, John. *Propitious Esculent: The Potato in World History* (2008), 315pp a standard scholarly history
- Salaman, Redcliffe N. (1989) [1949. *The History and Social Influence of the Potato*, Cambridge University Press.
- Spooner, David M.; McLean, Karen; Ramsay, Gavin; Waugh, Robbie; Bryan, Glenn J. (October 2005). "A single domestication for potato based on multilocus amplified fragment length polymorphism genotyping" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1253605). *Proceedings of the National Academy of Sciences of the United States of America*. **102** (41). National Academy of Sciences of the United States of America: 14694–14699. Bibcode:2005PNAS..10214694S (https://ui.adsabs.harvard.edu/abs/2005PNAS..10214694S). doi:10.1073/pnas.0507400102 (https://doi.org/10.1073%2Fpnas.0507400102). PMC 1253605 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1253605). PMID 16203994 (https://pubmed.ncbi.nlm.nih.gov/16203994).
- Stevenson, W.R., Loria, R., Franc, G.D., and Weingartner, D.P. (2001) *Compendium of Potato Diseases*, 2nd ed, Amer. Phytopathological Society, St. Paul, MN.
- The World Potato Atlas (https://www.worldpotatoatlas.com/), released by the International Potato Center in 2006 and regularly updated.
- World Geography of the Potato at UGA.edu (https://web.archive.org/web/20060604061759/http://www.lanra.uga.edu/potato/), released in 1993.
- Zuckerman, Larry. *The Potato: How the Humble Spud Rescued the Western World.* (1998). 304 pp. Douglas & McIntyre. ISBN 0-86547-578-4.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Potato&oldid=1240606982"